GOOD MORNING!
早上好!
안녕하세요!

DAY 6

DAY 5 (FINAL PROJECT)

- 비즈니스/System 요구 사항 업데이트
- 시스템 설계 및 프로세스 정립
  - System(High Level) Design
  - Schedule/Time Management
- 역할 분담 및 일정 조율
- 개발 환경 구축(맵 디자인, SW 개발,문서 통합 관리)
- 멀티 로봇 환경 구축 및 네비게이션
- 멀티 로봇 개별 업무 수행
- 멀티 로봇 협동 업무 수행
- (Optional)Turtlebot4 각종 센서 데이터의 이해와 적용

DAY 6 (FINAL PROJECT)

- 시스템 설계에 기반한 객체 감지 모델 구현
- 로봇 환경에 적용 및 Unit Test
- 모듈로 제작하고 launch파일로 구현
- code 정리 및 버전관리, 문서 작성 및 영상 촬영, 팀 내 기술 브리핑

- 시스템 설계에 기반한 SysMon 설계 구현
- 로봇 환경에 적용 및 Unit Test
- 모듈로 제작하고 launch파일로 구현
- code 정리 및 버전관리, 문서 작성 및 영상 촬영, 팀 내 기술 브리핑

DAY 7 (FINAL PROJECT)

- 시스템 설계에 기반한 AMR 제어 구현
- 로봇 환경에 적용 및 Unit Test
- 모듈로 제작하고 launch 파일로 구현
- code 정리 및 버전관리, 문서 작성 및 영상 촬영, 팀 내 기술 브리핑

프로젝트 RULE NUMBER ONE!!!

Have Fun Fun Fun!

PROJECT SCHEDULE UPDATE BY EACH TEAM

BUSINESS/SOLUTION REQUIREMENT UPDATE BY EACH TEAM

Using the posted notes and flipchart as needed

SYSTEM REQUIREMENT UPDATE BY EACH TEAM

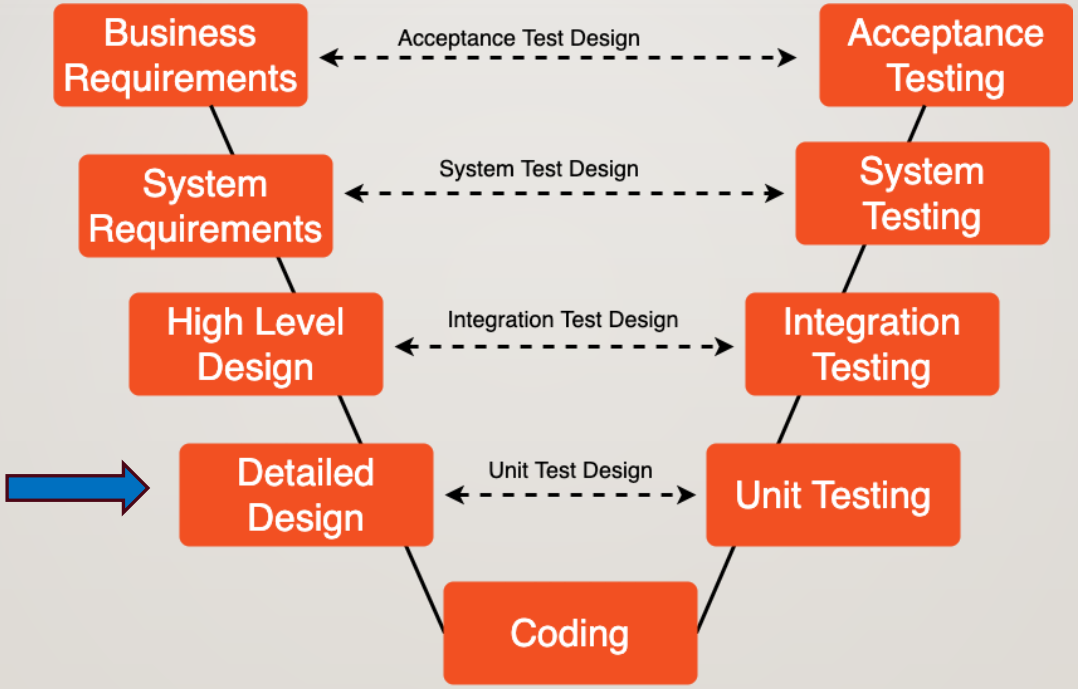
Using the posted notes and flipchart as needed

SYSTEM DESIGN ARCHITECTURE UPDATE BY EACH TEAM

Using the posted notes and flipchart as needed

SYSTEM DESIGN ARCHITECTURE PRESENTATION SHOULD BE DONE HERE BUT POSTPONED TO DAY 7 AFTERNOON!!!!

SPRINT 1

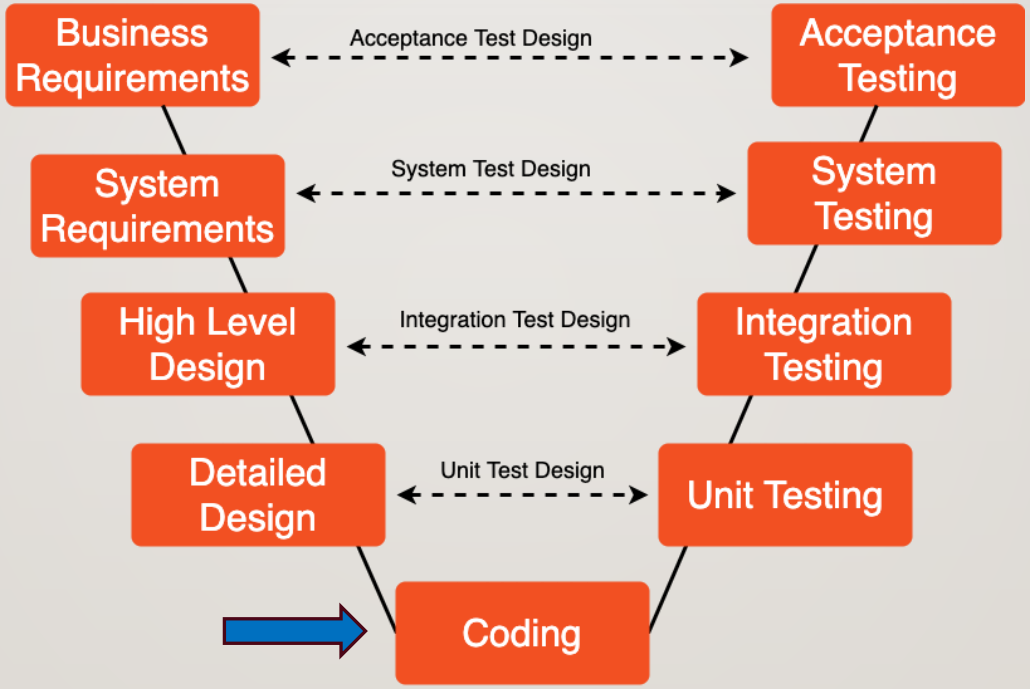


SDLC - V Model - notepub.io

TEAM EXERCISE 12

Perform Detail Design of Sprint 1 using Process Flow Diagram

SPRINT 1



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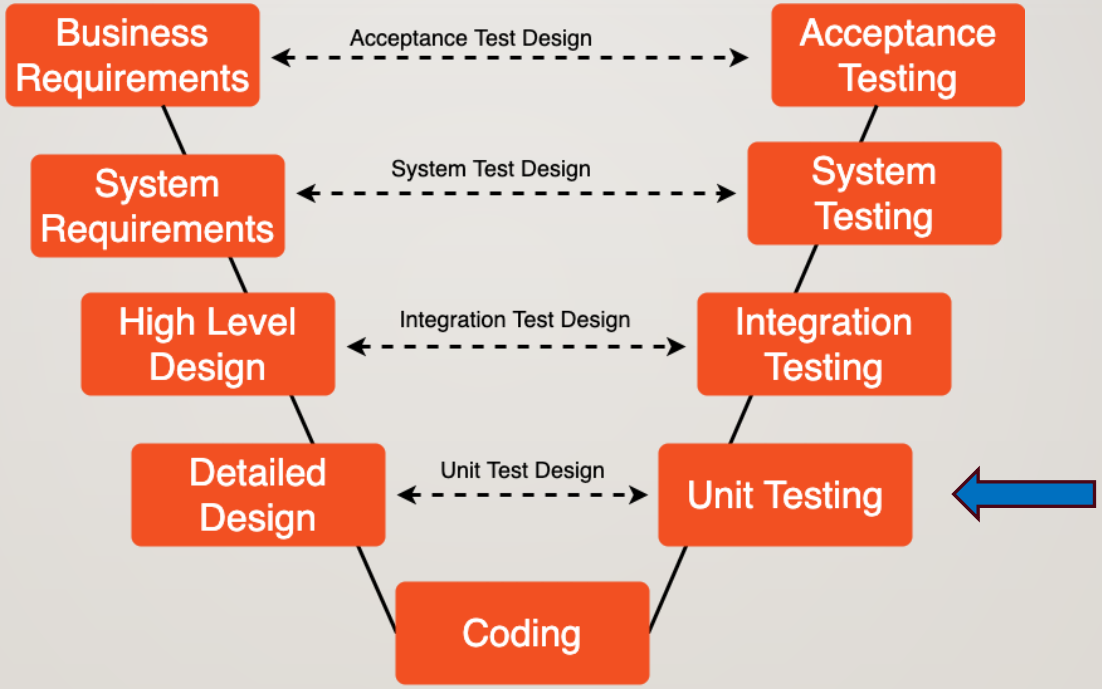
SPRINT 1



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TEAM EXERCISE 13

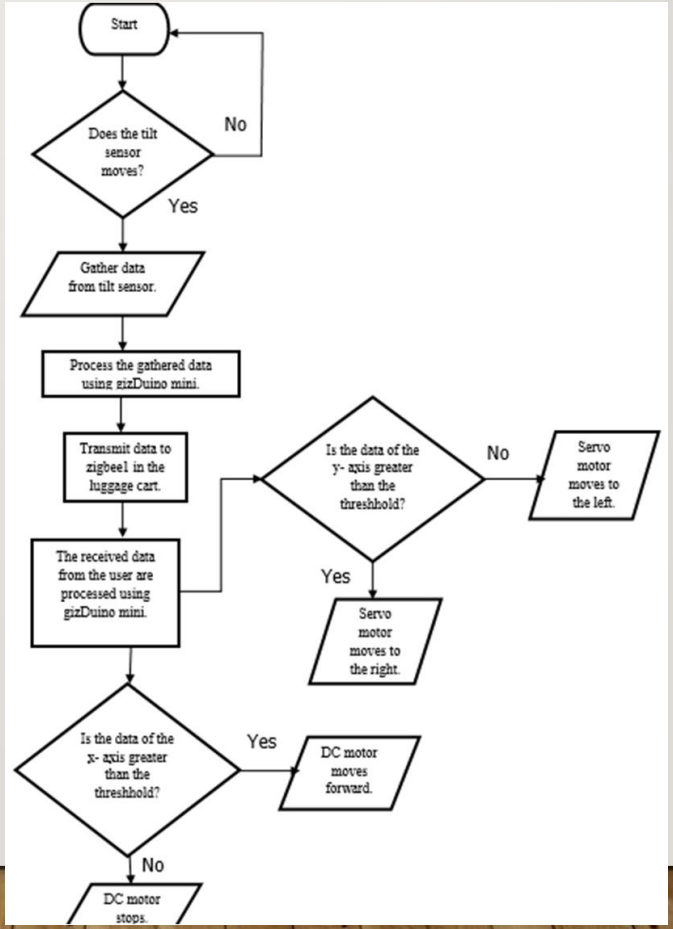
Perform coding and testing of Sprint 1 Module

(평가 2) DESIGN/CODE/TEST REVIEW BY EACH TEAM

Show actual results against the expected results and explain the code written

WHAT TO DO IN CODE REVIEW

- Explain
  - Design



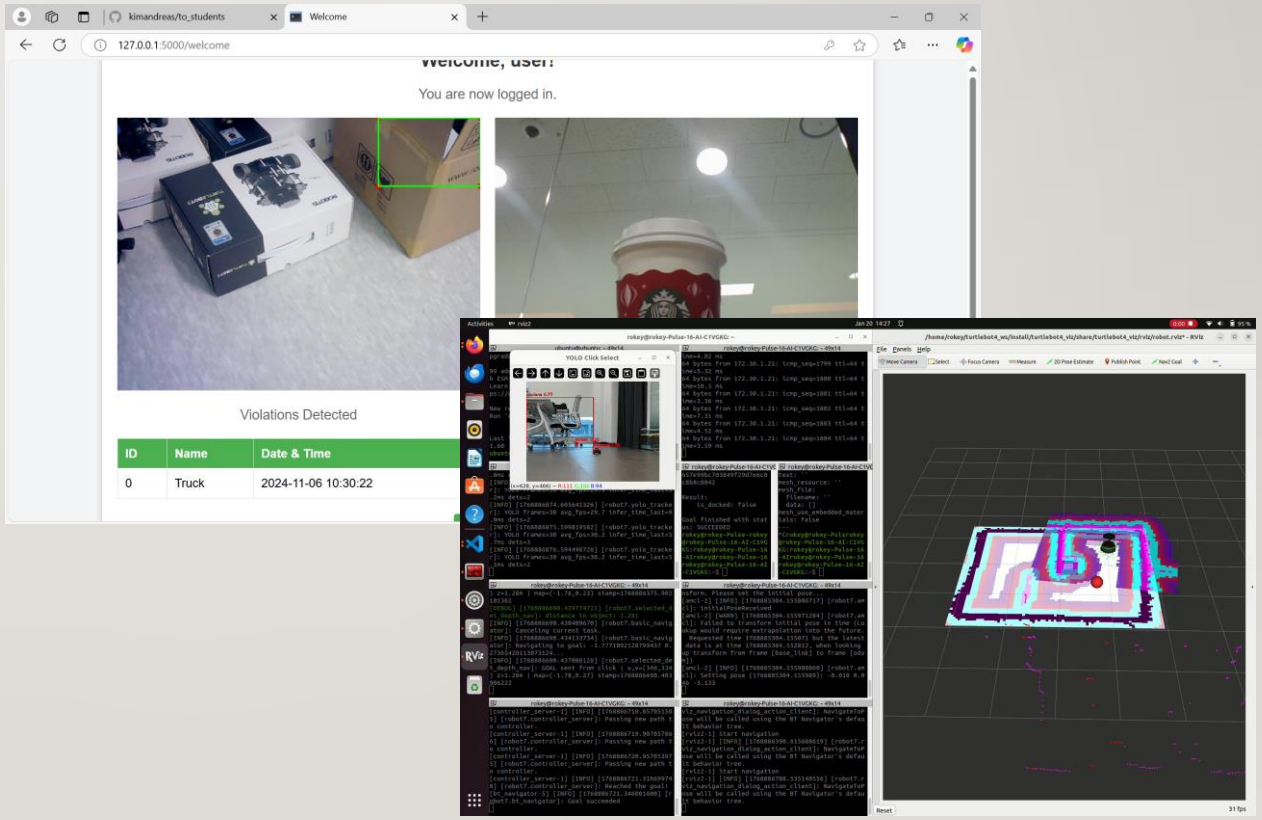
WHAT TO DO IN CODE REVIEW

- Explain
  - Design
  - Code

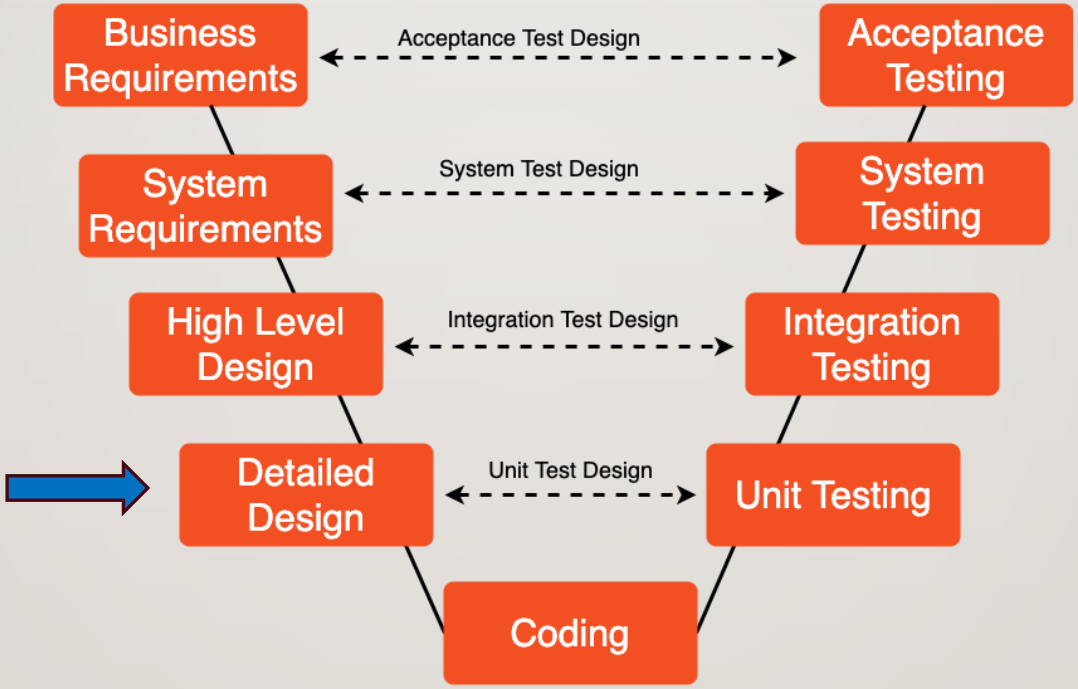
```python
def decode_compressed_depth_16uc1(msg: CompressedImage) -> Tuple[Optional[np.ndarray], str]:
    depth = cv2.imdecode(png, cv2.IMREAD_UNCHANGED)

    if depth is None:
        return None, f"cv2.imdecode failed (png_offset={i} len={len(buf)} format='{msg.format}')"

    # Normalize shape: want HxW
    if depth.ndim == 3:
        depth = depth[:, :, 0]

    if depth.dtype not in (np.uint16, np.float32):
        return None, f"decoded dtype={depth.dtype} (expected uint16 or float32) png_offset={i}"

    return depth, f"ok png_offset={i} shape={depth.shape} dtype={depth.dtype}"


class SelectedDetDepthNav(Node):
    """Time-sync selected detections with depth, then send Nav2 goals + marker."""

    def __init__(self):
        super().__init__("selected_det_depth_nav")

        # ----------------------------
        # Topic setup (namespace-aware)
        # ----------------------------
        # detection topic is relative so it follows __ns
        self.det_topic = "selected_detection"

        ns = self.get_namespace().rstrip("/")
        self.depth_topic = f"{ns}/oakd/stereo/image_raw/compressedDepth"
        self.info_topic = f"{ns}/oakd/rgb/camera_info"

        self.get_logger().info(f"Detection topic : {self.resolve_topic_name(self.det_topic)}")
        self.get_logger().info(f"Depth topic     : {self.depth_topic}")
```

WHAT TO DO IN CODE REVIEW

- Explain
  - Design
  - Code
  - Test/Validation
    - Test Code
    - Log Output
    - Screenshot/video
    - Photo/Video
    - ….



SPRINT 2



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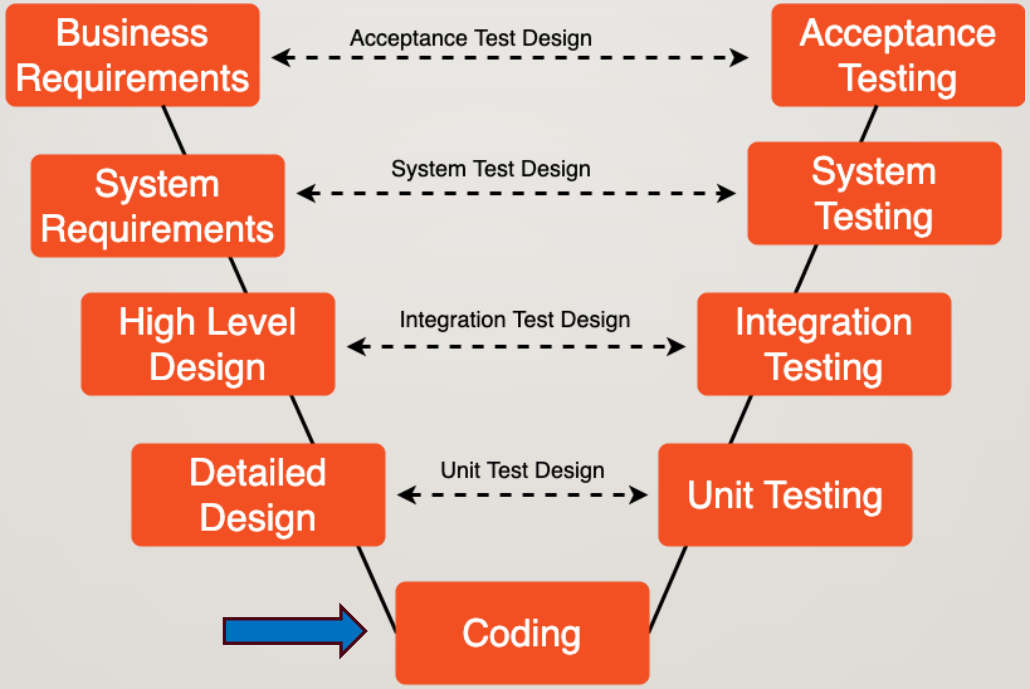
TEAM EXERCISE 14

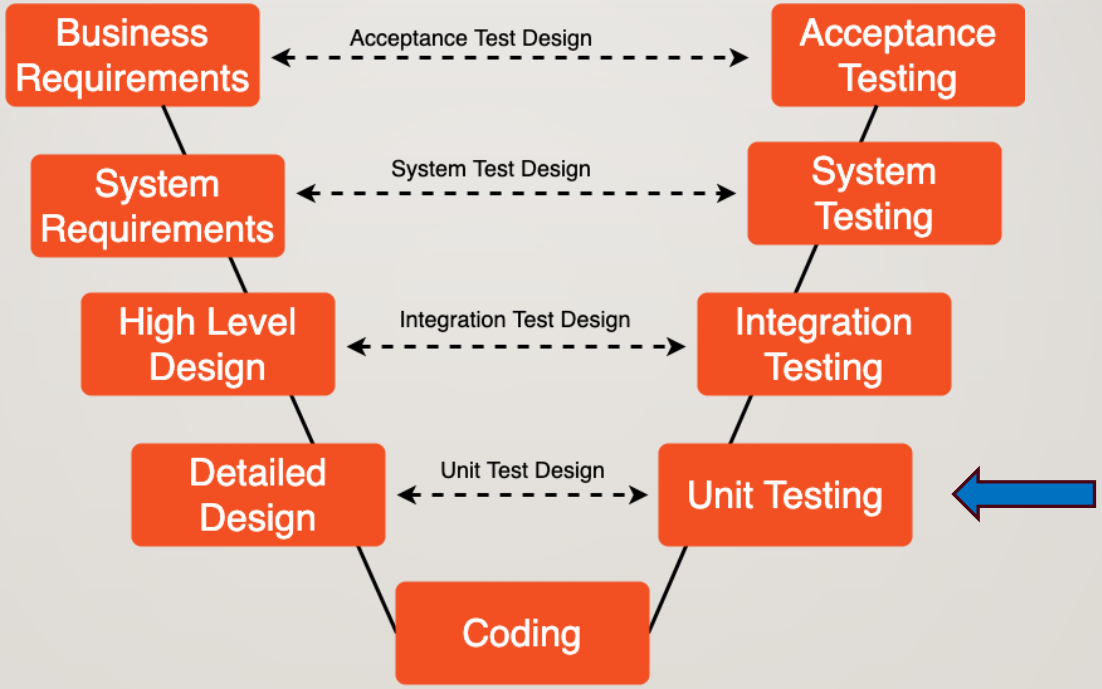
Perform Detail Design of Sprint 2 using Process Flow Diagram

SPRINT 2



SDLC - V Model - notepub.io

SPRINT 2



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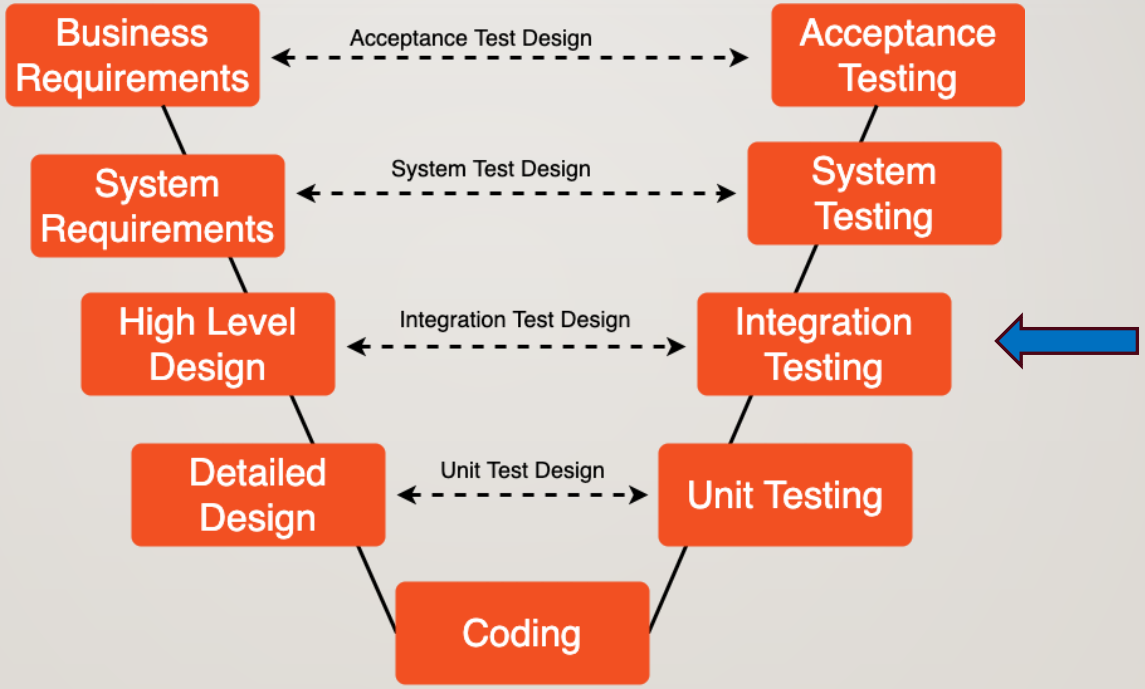
TEAM EXERCISE 15

Perform coding and testing of Sprint 2 Module

(평가 3) DESIGN/CODE/TEST REVIEW BY EACH TEAM

Show actual results against the expected results and explain the code written

SPRINT 1&2 – INTEGRATION & TEST



SDLC - V Model - notepub.io

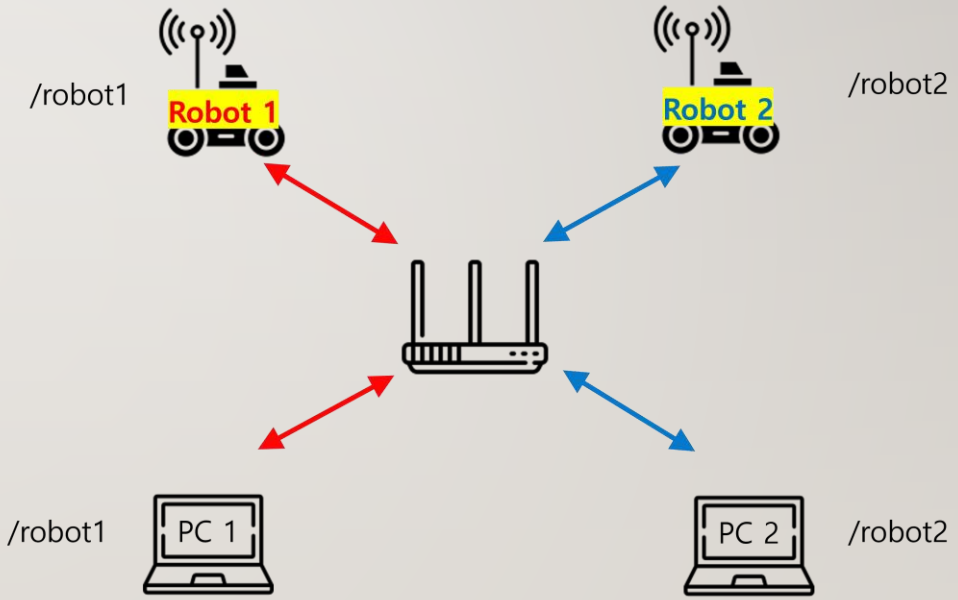
TEAM EXERCISE 16

Perform integrate and test of Sprint 1&2 Modules

SETTING SOLUTION NETWORK

DAY 5 - System Monitor & Multi Robot

Aa 이름	상태
Multi Robot Standard Setup	시작 전
Multi Robot Custom Discovery Setup	시작 전



프로젝트 RULE NUMBER ONE!!!

Are we having Fun???